12. Fragments (Sub-Activity)

Android Studio

Introduction to Fragments | Android

Fragment is a piece of an activity that enables a more modular activity design.

A fragment encapsulates functionality so that it is easier to reuse within activities and layouts.

Android devices exist in a variety of screen sizes and densities.

Fragments simplify the reuse of components in different layouts and their logic.

You can build single-pane layouts for handsets (phones) and multi-pane layouts for tablets.

You can also use fragments also to support different layouts for landscape and portrait orientation on a smartphone.

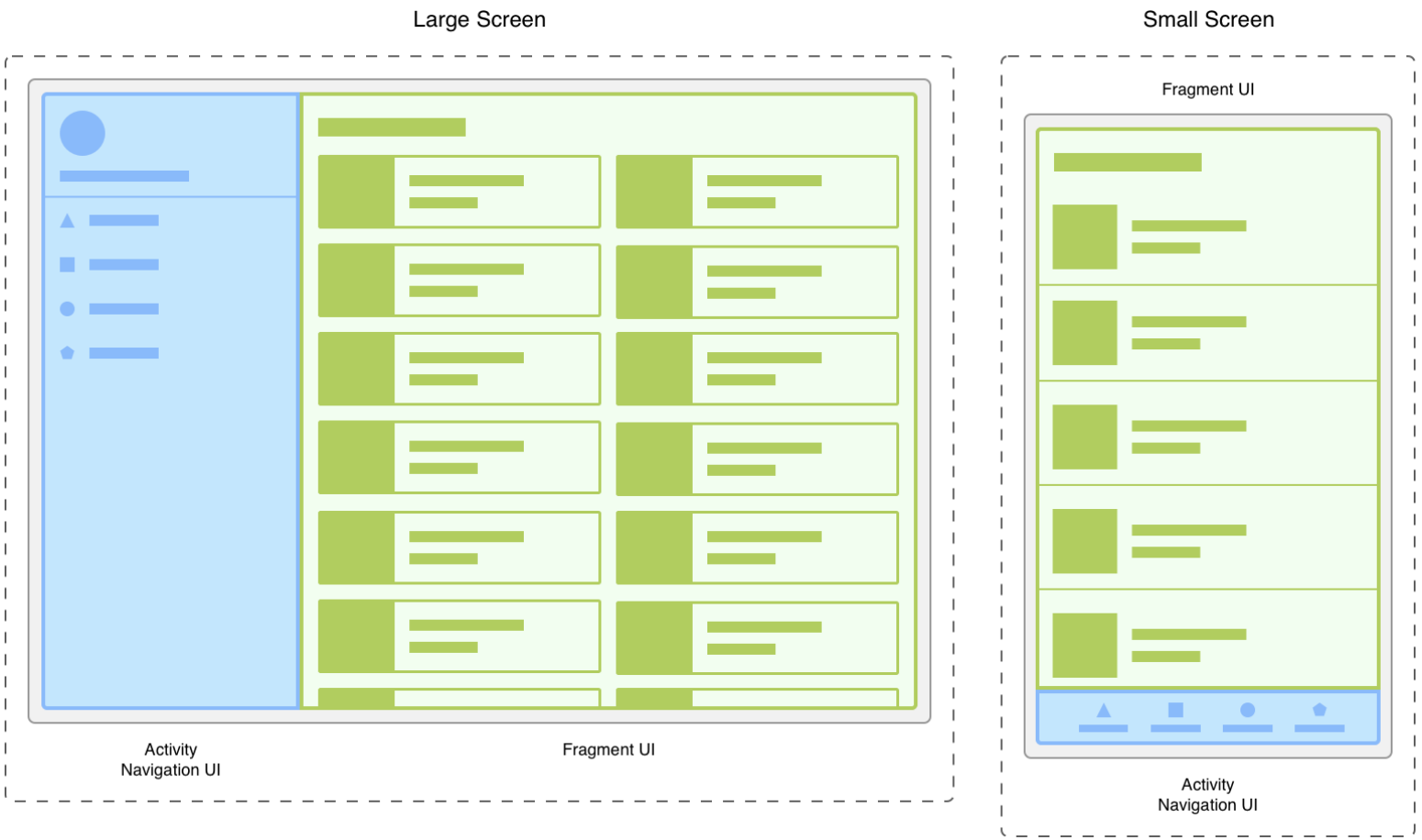
- في الأندرويد الـ Fragment هي جزء من الـ Activity و يمكننا القول بأنها Sub-Activity.
- هذه الـ Fragment هي تمثل جزء من واجهة المستخدم في الـ Activity الحالية.
- نستطيع مثلا دمج 3 من الـ Fragments داخل Activity.
- و لا ننسى أن دورة حياة ال Fragment تتأثر بدورة حياة الـ Activity.
- قبل استخدام الـ Fragments كنا نستطيع إظهار Activity واحدة فقط على الشاشة في الوقت الحالي للمستخدم,
- إذن كان من الصعب تقسيم الشاشة إلى عدد من الأجزاء كل جزء يؤدي مهمة معينة.
- و لكن مع الـ Fragments استطعنا عمل ذلك بتقسيم الـ Activity إلى عدد من الأجزاء Fragments و التحكم في كل جزء بشكل منفصل.
- بذلك كل Fragment لها events و Layouts خاصة بها مثل الـ Activity تمام.

ميزات Fragment

- نستطيع إضافة و حذف الـ Fragment أثناء تشغيل الـ Activity.
- الـ Fragment لها Layout خاصة بها و دورة حياة مستقلة كذلك
- الـ Fragment يمكن اعادة استخدامها في عدة Activities.
- يمكننا أيضاً دمج عدد من الـ Fragment داخل Activity واحدة.
- يمكن أن الـ Fragment تشكّل جزء من الـ Activity و ممكن تأخذ الـ Activity كاملة.
- ممكن إظهار عدد من الـ Fragment في نفس الوقت و ممكن عمل تبديل swapping بينها.

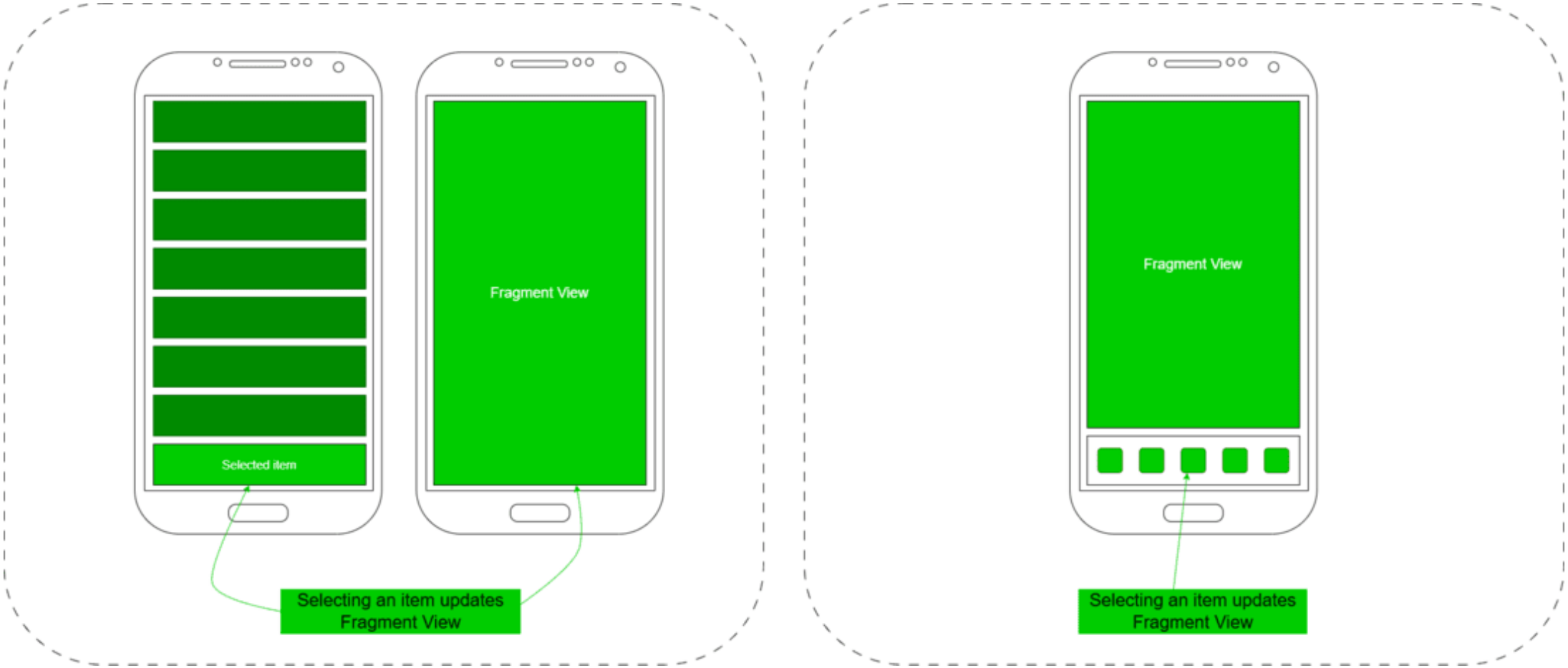
Two versions of the same screen on different screen sizes.

On the left, a large screen contains a navigation drawer that is controlled by the activity and a grid list that is controlled by the fragment.



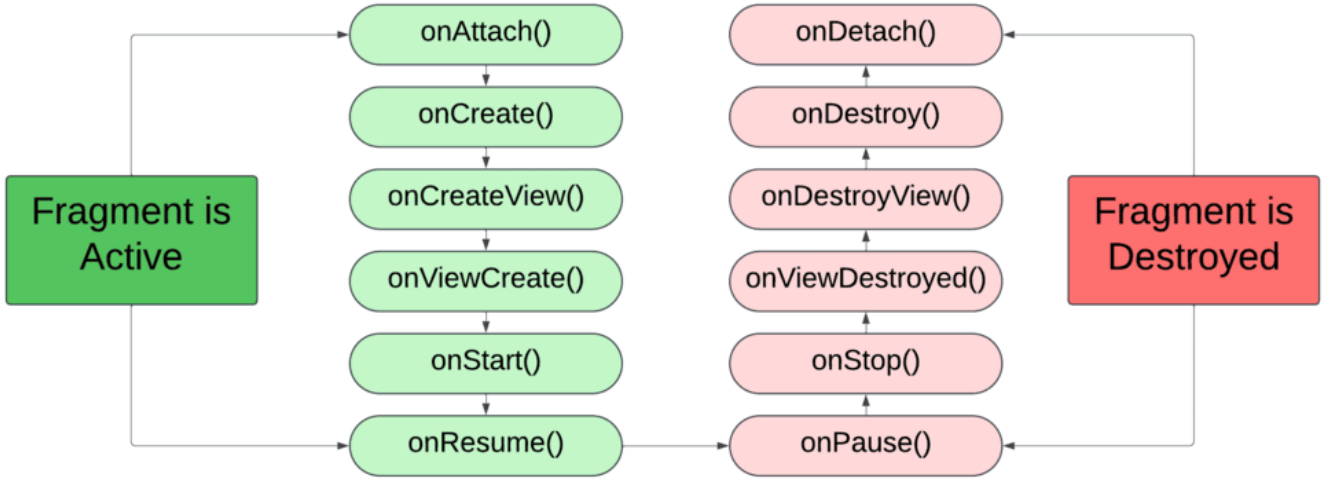
On the right, a small screen contains a bottom navigation bar that is controlled by the activity and a linear list that is controlled by the fragment.

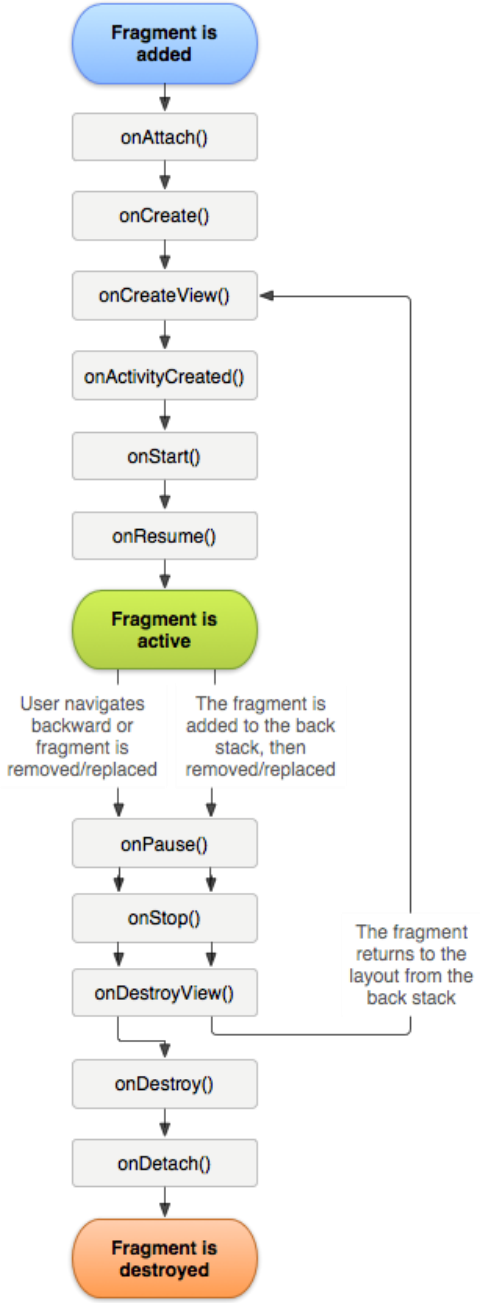
Image shows the use cases of fragments through navigations.



Fragment Lifecycle

Android fragments have their own lifecycle very similar to an android activity.





Fragment Lifecycle

Android fragmeonAttach() : The fragment instance is associated with an activity instance. The fragment and the activity is not fully initialized. Typically you get in this method a reference to the activity which uses the fragment for further initialization work.

onCreate() : The system calls this method when creating the fragment. You should initialize essential components of the fragment that you want to retain when the fragment is paused or stopped, then resumed.

onCreateView() : The system calls this callback when it's time for the fragment to draw its user interface for the first time. To draw a UI for your fragment, you must return a View component from this method that is the root of your fragment's layout. You can return null if the fragment does not provide a UI.

onActivityCreated() : The onActivityCreated() is called after the onCreateView() method when the host activity is created. Activity and fragment instance have been created as well as the view hierarchy of the activity. At this point, view can be accessed with the findViewById() method. example. In this method you can instantiate objects which require a Context object

onStart() : The onStart() method is called once the fragment gets visible.

onResume() : Fragment becomes active.

onPause() : The system calls this method as the first indication that the user is leaving the fragment. This is usually where you should commit any changes that should be persisted beyond the current user session.

onStop() : Fragment going to be stopped by calling onStop()

onDestroyView() : Fragment view will destroy after call this method

onDestroy() :called to do final clean up of the fragment's state but Not guaranteed to be called by the Android platform.

The following lifecycle events come into play when you add a fragment:

- `onAttach`: The fragment attaches to its host activity.
- `onCreate`: A new fragment instance initializes, which always happens after it attaches to the host. Fragments are a bit like viruses. :]
- `onCreateView`: A fragment creates its portion of the view hierarchy, which is added to its activity's view hierarchy.
- `onActivityCreated`: The fragment's activity has finished its own `onCreate` event.
- `onStart`: The fragment is visible. A fragment starts only after its activity starts and often starts immediately after its activity does.
- `onResume`: The fragment is visible and interactable. A fragment resumes only after its activity resumes and often resumes immediately after the activity does.

But wait, the fragment isn't done! These lifecycle events happen when you remove a fragment:

- `onPause`: The fragment is no longer interactable. This occurs when either the fragment is about to be removed or replaced or when the fragment's activity pauses.
- `onStop`: The fragment is no longer visible. This occurs either after the fragment is about to be removed or replaced or when the fragment's activity stops.
- `onDestroyView`: The view and related resources created in `onCreateView` are removed from the activity's view hierarchy and destroyed.
- `onDestroy`: The fragment does its final clean up.
- `onDetach`: The fragment is detached from its activity.

As you can see, the fragment's lifecycle intertwines with the activity's lifecycle. But it has extra events that are particular to the fragment's view hierarchy, state and attachment to its activity.

Types of Fragments

Single frame fragments : Single frame fragments are using for hand hold devices like mobiles, here we can show only one fragment as a view.

List fragments : fragments having special list view is called as list fragment

Fragments transaction : Using with fragment transaction. we can move one fragment to another fragment.

Handling the Fragment Lifecycle

A Fragment exist in three states:

Resumed : The fragment is visible in the running activity.

Paused : Another activity is in the foreground and has focus, but the activity in which this fragment lives is still visible (the foreground activity is partially transparent or doesn't cover the entire screen).

Stopped : The fragment is not visible. Either the host activity has been stopped or the fragment has been removed from the activity but added to the back stack. A stopped fragment is still alive (all state and member information is retained by the system). However, it is no longer visible to the user and will be killed if the activity is killed.

Defining and using fragments

- https://www.geeksforgeeks.org/android/introduction-fragments-android/

Create Fragment : Navigation By Buttons

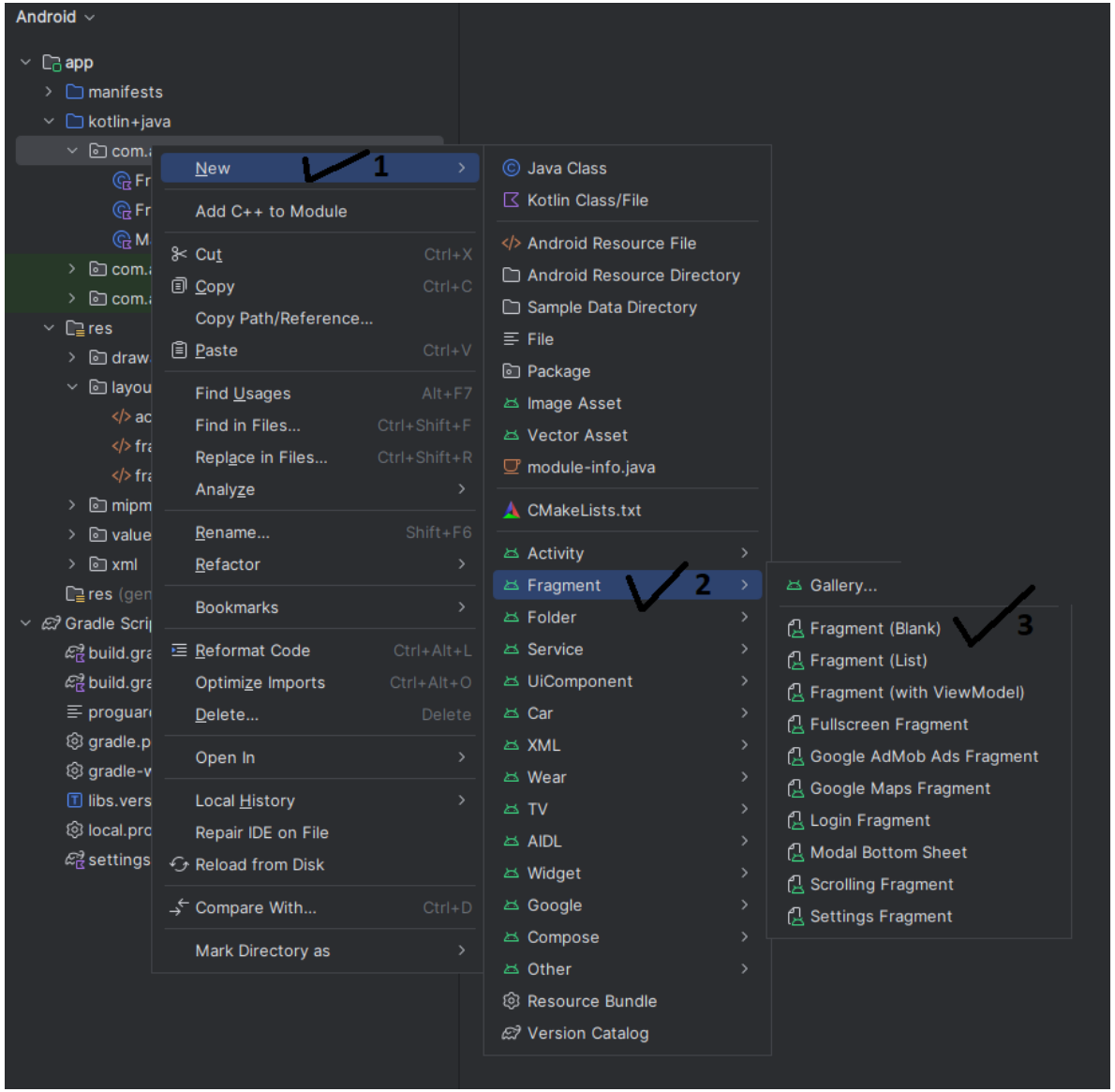
Project name: ***_CreateFragment1

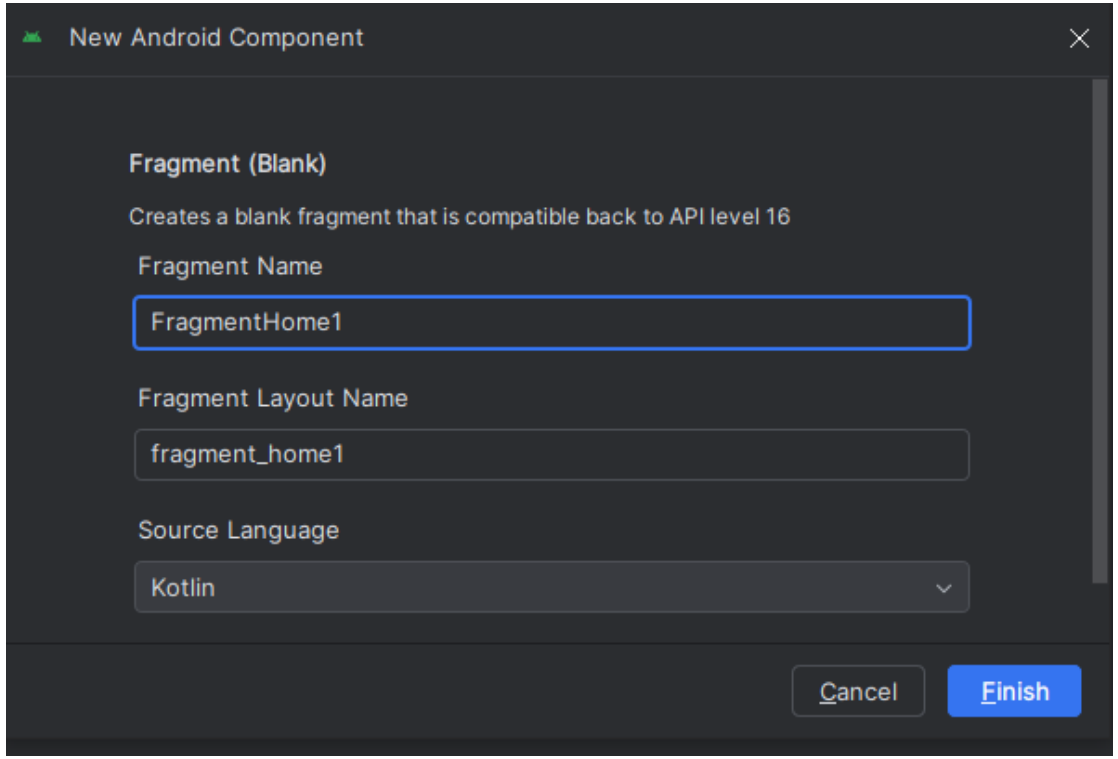
Create Two Fragments:

Fragment1, Fragment2

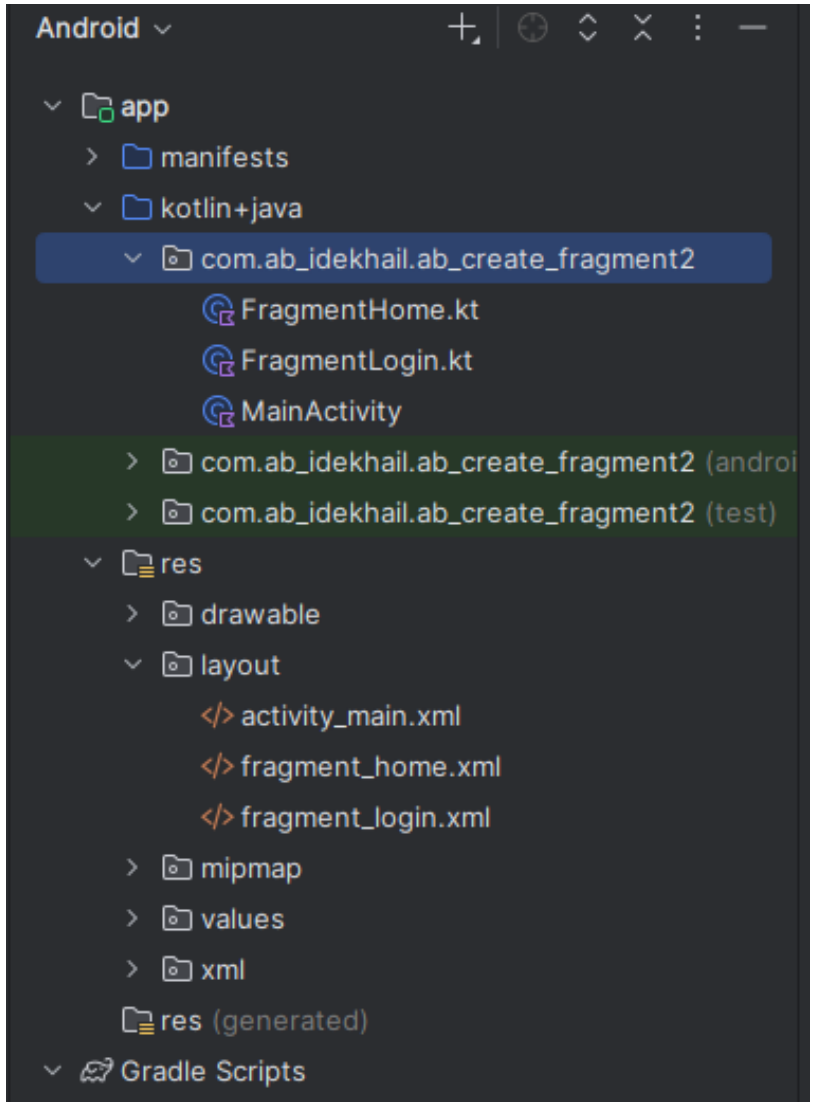
MainActivity

Create New Fragment





Design fragment_Home




```xml
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@color/black"
    tools:context=".FragmentHome">

    <!-- TODO: Update blank fragment layout -->
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:padding="20dp"
        android:layout_gravity="center"
        android:background="#AABBCC"
        android:textColor="@color/black"
        android:textSize="25sp"
        android:text="Fragment Home" />

</FrameLayout>
```

FragmentHome Programming

```
3   import android.os.Bundle
4   import androidx.fragment.app.Fragment
5   import android.view.LayoutInflater
6   import android.view.View
7   import android.view.ViewGroup
    4 Usages
8   class FragmentHome : Fragment() {
9       override fun onCreate(savedInstanceState: Bundle?) {
10          super.onCreate(savedInstanceState)
11          arguments?.let {
12              //
13          }
14      }
15      override fun onCreateView(
16          inflater: LayoutInflater, container: ViewGroup?,
17          savedInstanceState: Bundle?
18      ): View? {
19          return inflater.inflate( resource = R.layout.fragment_home,  root = container,  attachToRoot = false)
20      }
21  }
```

Activity_main Design

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <androidx.fragment.app.FragmentContainerView
        android:id="@+id/fragmentContainerView"
        android:name="com.ab_idekhail.ab_create_fragment2.FragmentHome"
        android:layout_width="match_parent"
        android:layout_height="600dp"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
```

```xml
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:gravity="center"
        android:layout_margin="10dp"
        android:padding="10dp"
        android:background="#999"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/fragmentContainerView">

        <Button
            android:id="@+id/fragmentH"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_margin="15dp"
            android:layout_weight="1"
            android:text="Fragment Home" />

        <Button
            android:id="@+id/fragmentL"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_margin="15dp"
            android:layout_weight="1"
            android:text="Fragment Login"
            app:layout_constraintBottom_toBottomOf="parent" />
    </LinearLayout>

</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.kt

```kotlin
class MainActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        enableEdgeToEdge()
        setContentView(R.layout.activity_main)
        val fragmentH = findViewById<Button>( id = R.id.fragmentH)
        val fragmentL = findViewById<Button>( id = R.id.fragmentL)
        if (savedInstanceState == null) {
            supportFragmentManager.beginTransaction()
                .add( containerViewId = R.id.fragmentContainerView, FragmentHome())
                .commit()
        }
        fragmentH.setOnClickListener {
            supportFragmentManager.beginTransaction()
                .add( containerViewId = R.id.fragmentContainerView, FragmentHome())
                .commit()
        }
        fragmentL.setOnClickListener {
            supportFragmentManager.beginTransaction()
                .add( containerViewId = R.id.fragmentContainerView, FragmentLogin())
                .commit()
        }
    }
}
```

Creating a Tab Layout with Icons and Name: Using Fragments in Android Studio (Kotlin))

Project name: ASB2_CreateFragment.SecondActivity

Create Two Fragments:

Fragment1, Fragment2

```
28          buildFeatures { this: ApplicationBuildFeatures
29              dataBinding = true
30              viewBinding = true
31          }
32      }
33      compileOptions { this: CompileOptions
34          sourceCompatibility = JavaVersion.VERSION_1_8
35          targetCompatibility = JavaVersion.VERSION_1_8
36      }
37      kotlinOptions { this: KotlinJvmOptions
38          jvmTarget = "1.8"
39      }
40  }
41  dependencies { this: DependencyHandlerScope
42      implementation("androidx.core:core-ktx:1.13.1")
43      implementation("androidx.appcompat:appcompat:1.7.0")
44      implementation("com.google.android.material:material:1.12.0")
45      implementation("androidx.constraintlayout:constraintlayout:2.1.4")
46      testImplementation("junit:junit:4.13.2")
47      androidTestImplementation("androidx.test.ext:junit:1.1.5")
48      androidTestImplementation("androidx.test.espresso:espresso-core:3.5.1")
49      implementation("com.google.android.material:material:1.12.0")
50  }
```

https://github.com/idekhail/ASB2_CreateFragment

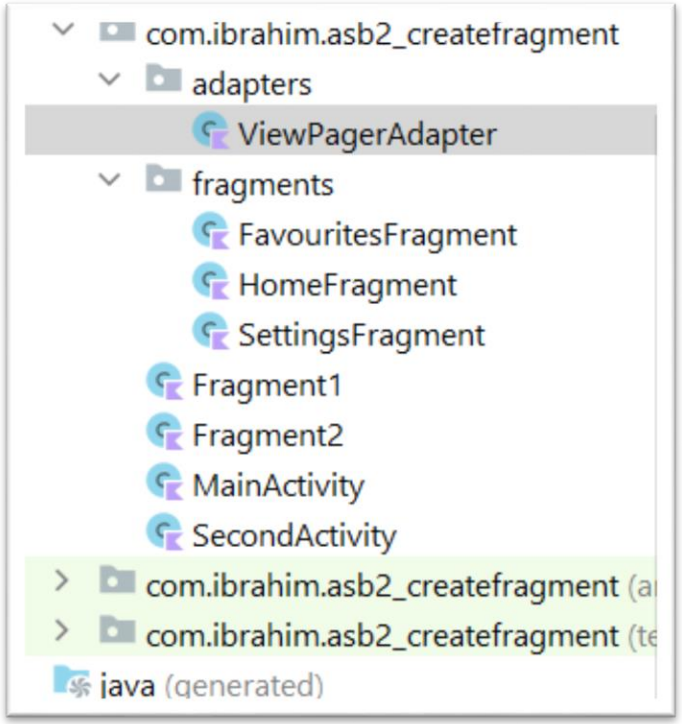
Project name: ASB2_CreateFragment.SecondActivity

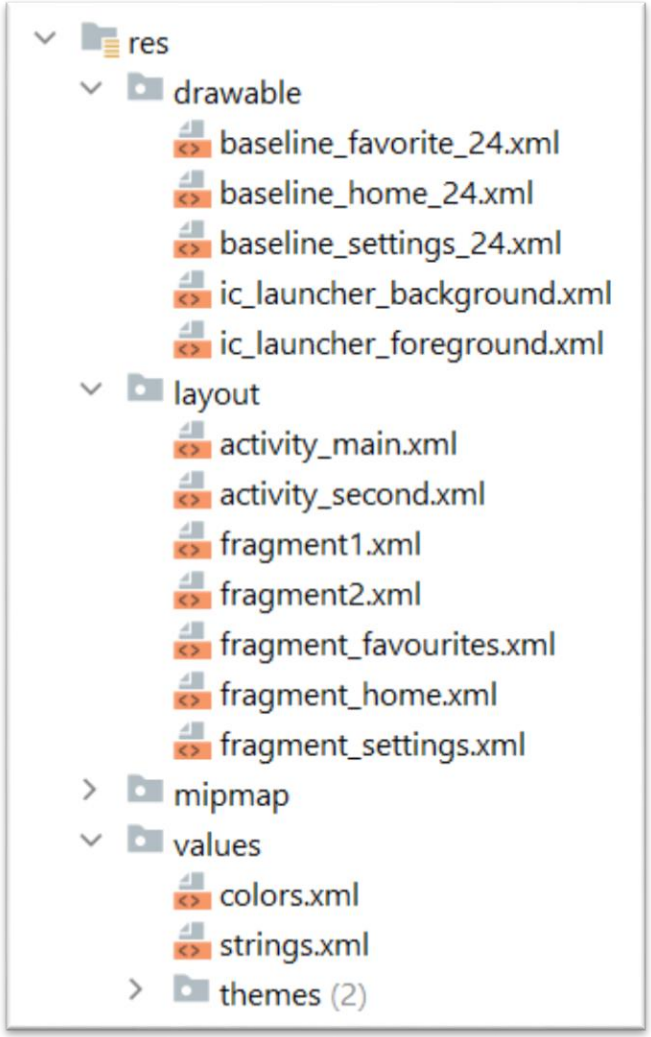
Create Two Fragments:

HomeFragment,

FavoritesFragment

SettingFragment





https://github.com/idekhail/ASB2_CreateFragment